

25 #5: 1, 6, 9, 11, 12

26 #5: 1, 2, 4

Homework 5, math 3210

5 #1: An example of a nested sequence of bounded open intervals that does not have a point in its intersection.

$$I_m \subset I_n \quad \forall n, m \in \mathbb{N} \text{ s.t. } \bigcap I_n = \emptyset.$$

An example is: $(0, \frac{1}{n+1}) \subset (0, \frac{1}{n})$ and for all n , $\bigcap I_n = \emptyset$.

6: Find convergent subsequences for:

a) $a_n = (-1)^n$

$$(-1)^k \text{ where } k = 2n \\ = (1, 1, 1, 1, \dots) \rightarrow 1$$

$$\lim(a_n) = |(-1)^{2n} - 1| = |1 - 1| = 0 < \epsilon \text{ for all } n \in \mathbb{N}.$$

b) $a_n = \sin\left(\frac{n\pi}{4}\right)$

$$a_k = \sin\left(\frac{k\pi}{4}\right) \text{ where } k = 4n \\ = (0, 0, 0, \dots) \rightarrow 0$$

$$\lim(a_k) = \left| \sin\left(\frac{4n\pi}{4}\right) - 0 \right| = |\sin(n\pi) - 0| = |0 - 0| = 0 < \epsilon \text{ for all } n \in \mathbb{N}.$$

c) $a_n = \frac{n}{2^{k_n}} - 1$, with k_n the largest integer k s.t. $2^k \leq n$.

Select the subsequence where " n " is a power of 2.

$$\text{Then } a_2 = \frac{n}{2^{k_n}} - 1, \quad n \in \{2^k\} \subset \mathbb{N}$$

$$\text{Then for all } k_n, \frac{n}{2^{k_n}} = 1 \text{ by def. of } k_n.$$

$$\text{So } a_2 = 1 - 1 = 0. \quad (0, 0, 0, 0, \dots) \rightarrow 0$$

$$\lim(a_2) = \left| \frac{n}{2^{k_n}} - 1 - 0 \right| = \left| \frac{2^n}{2^{k_n}} - 1 \right| = |1 - 1| = 0 < \epsilon \\ \text{for all } n \in \{2^k\}$$

2.5 #s: 9, 11, 12

2.6 #s: 1, 2, 4

Homework 5, cont.

9: Prove that a sequence which satisfies $|a_{n+1} - a_n| < 2^{-n}$ for all n is a Cauchy sequence.

Let $m = n+1$ and $n = n$. A sequence satisfying $|a_m - a_n| < 2^{-n}$ is Cauchy if for any $\varepsilon > 0$, there is an N such that

$$|a_m - a_n| < \varepsilon \quad \text{for all } n, m > N$$

Sketch: $|a_m - a_n| < \frac{1}{2^n} < \varepsilon$; $\frac{1}{\varepsilon} < 2^n$; $\log_2\left(\frac{1}{\varepsilon}\right) < n$
proof: Suppose $m = n+1$. Then select $N = \log_2\left(\frac{1}{\varepsilon}\right)$. For all $\varepsilon > 0$, then $|a_m - a_n| < 2^{-n} < \varepsilon \Rightarrow n > \log_2\left(\frac{1}{\varepsilon}\right) = N$, which holds for all n, m because $m > n$.

11: Let $S_n = \sum_{k=1}^n \frac{1}{k2^k}$ be the sequence of partial sums of the series $\sum_{k=1}^{\infty} \frac{1}{k2^k}$. Prove that $\{S_n\}$ converges.

$$S_n = \left(\frac{1}{2}, \frac{1}{2} + \frac{1}{8}, \frac{1}{2} + \frac{1}{8} + \frac{1}{24}, \frac{1}{2} + \frac{1}{8} + \frac{1}{24} + \frac{1}{64}, \dots\right)$$

$$\text{for } m > n \text{ we have } |S_m - S_n| = \left| \sum_{k=n+1}^m \frac{1}{k2^k} \right| = \sum_{k=n+1}^m \frac{1}{k2^k} \\ \leq \sum_{k=n+1}^m \frac{1}{2^k} \leq \frac{1}{2^{n+1}} \sum_{k=0}^{\infty} \frac{1}{2^k} = \frac{1}{2^n}$$

$$\text{Using the fact that } \sum_{k=0}^{\infty} \frac{1}{2^k} = 2.$$

Since $\lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$ by Ex. 2.35, given $\varepsilon > 0$, there is an N such that $n > N$ implies $\frac{1}{2^n} < \varepsilon$. Then $|S_m - S_n| < \varepsilon$ for all n, m with $m > n > N$. This means that S_n is Cauchy and therefore converges.

2.5 Hs: 12

2.6 Hs: 1, 2, 4

Homework 5, cont.

12. Given a series $\sum_{k=1}^{\infty} a_k$, set $S_n = \sum_{k=1}^n a_k$ and $T_n = \sum_{k=1}^n |a_k|$. Prove that $\{S_n\}$ converges if $\{T_n\}$ is bounded.

$$S_n = (a_1, a_1 + a_2, a_1 + a_2 + a_3, \dots, a_1 + a_2 + \dots + a_n)$$

$$T_n = (|a_1|, |a_1| + |a_2|, |a_1| + |a_2| + |a_3|, \dots, |a_1| + |a_2| + |a_3| + \dots + |a_n|)$$

Since $|a_n| \geq 0$ for all $n \in \mathbb{N}$, $|a_1| + |a_2| + \dots + |a_n| \geq 0$.

So 0 is a lower bound for T_n .

Likewise, $t_1 \leq t_2 \leq t_3 \leq \dots \leq t_n$ by the same fact as above.

So, $T_n = |a_1| + |a_2| + \dots + |a_n|$ is an upper bound for t_n .

Thus, $0 \leq t_n \leq |a_1| + |a_2| + \dots + |a_n|$.

Since T_n is bounded and monotone, T_n converges by MCT.

Thus, t_n is Cauchy.

Therefore, $\exists \epsilon > 0$ - + for $m > n > N$, $|t_m - t_n| < \epsilon$.

Now, assume S_n were Cauchy. Then $|S_m - S_n| =$

$$|a_{n+1} + a_{n+2} + \dots + a_m| \leq |a_{n+1}| + |a_{n+2}| + \dots + |a_m| = |t_m - t_n| < \epsilon$$

So our assumption yields a true statement. Therefore the assumption that S_n is Cauchy is true. Thus,

S_n converges.

Also: $\epsilon > |t_m - t_n| = \overbrace{\left| \sum_{k=n+1}^m |a_k| \right|}^{\text{Cauchy-ness}} \geq \left| \sum_{k=n+1}^m a_k \right| = |a_m - a_n|$

which implies Cauchy-ness of S_n .

2.6 #1: 1, 2, 4

2.6 #1: Find $\limsup a_n$ and $\liminf a_n$ for:

a) $a_n = (-1)^n = (-1, 1, -1, 1, -1, 1, \dots)$

$\liminf a_n = -1$

$\limsup a_n = 1$

b) $a_n = (-\frac{1}{n})^n = (-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}, \frac{1}{6}, \dots)$

$\liminf a_n = -1$

$\limsup a_n = \frac{1}{2}$

c) $a_n = \sin(n\pi/3) = ($

$\liminf a_n = -\sqrt{3}/2$

$\limsup a_n = \sqrt{3}/2$

#2: Find $\liminf$ and $\limsup$ of for the seq. of 2.5.6 (c).

$a_n = \frac{n}{2^{k_n}} - 1$ with k_n largest integer s.t. $2^{k_n} \leq n$.

$a_n = (0, 0, \frac{1}{2}, 0, \frac{1}{4}, \frac{1}{2}, \dots)$

Claim: $\liminf a_n = 0$ and $\limsup a_n = 1$.

1) let $i_n = \inf \{a_k : k \geq n\}$. (Claim $i_n = 0$ for all n . So, $i_n = \inf \{\frac{n}{2^{k_n}} - 1 : k_n \geq n\}$

≥ 0 . Since $2^{k_n} \leq n$ where $k_n = \max \{k \in \mathbb{Z} : 2^k \leq n\}$, $1 = \frac{n}{n} \leq \frac{n}{2^{k_n}}$

So, $\frac{n}{2^{k_n}} - 1 \geq \frac{n}{n} - 1 \geq 0$ for all n . is true.

Thus, $i_n = \inf \{a_k : k \geq n\} = 0$. This means $\liminf a_n = \lim i_n = 0$.

2) Let $s_n = \sup \{a_m : m \geq n\}$, then $\sup \{\frac{m}{2^{k_m}} - 1 : m \geq n\}$. Suppose $\frac{m}{2^{k_m}} \geq 2$. Then $m \geq 2 \cdot 2^{k_m} = 2^{k_m+1}$. This contradicts assumption $2^{k_m} \leq m \leq 2^{k_m+1}$.

Therefore, $\frac{m}{2^{k_m}} \leq 2$ and $\frac{m}{2^{k_m}} - 1 \leq 2 - 1 = 1$. So 1 is an upper bound.

Suppose F.S.O.C. that $\exists p < 1$ s.t. $p \geq a_m \forall m \geq n$. By Arch. Prop, $\exists m > \log_2(1/p)$. Note. $p < 1$, so $\log_2(1/p)$ is well defined. So,

$\frac{2^m}{2^{k_m}} = \frac{2^m - 1}{2^{k_m}} \geq p$ contradicts p being an upper bound.

So 1 is the best upper bound. So, $s_n = \sup \{a_m : m \geq n\} = 1$.

Then $\lim s_n = \limsup \{a_m : m \geq n\} = \lim 1 = 1$.

2.6 #4

Homework 5, cont.

4: If $\limsup a_n$ & $\limsup b_n$ are finite, prove that:

$$\limsup (a_n + b_n) \leq \limsup a_n + \limsup b_n.$$

$$a_n + b_n = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

$$S_{a_n + b_n} = \sup \{a_k + b_k : k \geq n\} = \sup \{a_k : k \geq n\} + \sup \{b_k : k \geq n\}$$

by 1.5.7 (c).

Since $a_k + b_k \in a_n + b_n$, then $\sup \{a_k + b_k : k \geq n\} = \sup \{a_k : k \geq n\} + \sup \{b_k : k \geq n\}$ implies $\sup \{a_n + b_n\} \leq \sup \{a_n\} + \sup \{b_n\}$ by 1.5.7 (e).

By MLI $\limsup$ preserves non-strict inequalities,

$$\limsup (a_n + b_n) \leq \limsup a_n + \limsup b_n. \quad \square$$

Concept: $S_{a_n + b_n} \leq S_{a_n} + S_{b_n}$ by virtue of 1.5.7 (c) and (e).

where $S_{a_n + b_n} = \sup \{a_n + b_n\}$ and since $a_n + b_n \in a_n + b_n$

then $\sup \{a_n + b_n\} \leq \sup \{a_n\} + \sup \{b_n\}$

$$\text{and } \limsup a_n + b_n \leq \limsup a_n + \limsup b_n.$$